

Nir Guttman, PhD

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Professional Experience

Research Fellow

2023 – Present

Monash University, Victoria, Australia

As a Research Fellow at Monash University, I conduct research on developing and applying statistical tools to address various problems in gravitational-wave physics. I actively contribute to the gravitational-wave research group led by Professors Eric Thrane and Paul Lasky. Additionally, I am a member of the LIGO Scientific Collaboration.

- Conduct fundamental research on Bayesian inference methodologies, leveraging in-depth knowledge of statistics.
- Serve as the primary contributor and maintainer of statistical software.
- Supervise and collaborate with undergraduate and postgraduate students to support their research efforts.
- Acting as a referee, evaluating manuscripts submitted to The Astrophysical Journal.

Algorithm Developer

2013 – 2023

HP Indigo, Israel

In my role as an Algorithm Developer at Indigo, I specialized in the development of software algorithms that facilitate the interface between Indigo's press hardware and software. The nature of this work necessitates a robust understanding of the underlying physics of Indigo's printing process, which is leveraged to produce high-quality software solutions that ensure printing press stability and accuracy. This work is undertaken in a multidisciplinary environment that includes physicists, chemists, mechanical engineers, and software engineers.

- Developed and researched complex algorithms, utilizing a deep understanding of color science, including physics modeling of light-matter interactions.
- Frequently utilized various machine learning methods for regression, classification, and clustering, and carried out image processing projects.
- Followed a work methodology that included concept innovation, proof of concept experimentation, implementation at the production level, product quality assessment, and customer support.
- Maintained a complex software infrastructure responsible for supporting Indigo's press' algorithmic tasks.
- Designed experiments and solved challenging problems given a restricted and noisy system, accounting for measurement device capabilities and limitations.
- Offered international customer support, which included occasional travel as Indigo's expert for diagnosing and resolving customer escalations.
- Frequently reported to senior management and occasionally presented and served as a referee at internal HP conferences. Filed HP internal patent applications and collaborated with international patent lawyers.

Accomplishments

- Conceptualized and developed an algorithm that differentiated Indigo's capabilities from its competitors, directly contributing to the company's sales growth.
- Received three awards in recognition of substantial contributions to Indigo's technology and business.

- Co-authored nine patents.

Physics Data Analyst

2007 – 2013

The BABAR Experiment, Stanford University

As a graduate student pursuing a Ph.D. at Tel Aviv University, I contributed to the BABAR Experiment hosted at the Stanford Linear Accelerator Center. My research primarily focused on conducting precise physics measurements through the utilization of advanced data analysis techniques. The BABAR Experiment generated a distinctive dataset that necessitated a profound comprehension of physics principles and a comprehensive understanding of the detector's complexities to ensure accurate analysis outcomes.

- Performed data analysis using advanced curve-fitting techniques and multivariate analysis to separate signal and background, including the usage of neural networks. Strong emphasis was given to data-driven methods, statistical procedures, and uncertainty estimations.
- Collaborated with an international team consisting of physicists from various institutes.
- An integral part of the work included writing detailed documentation, frequently presenting at seminars, and serving as an official representative of the BABAR Experiment at conferences.

Accomplishments

- Conducted a significant portion of research while serving as a visiting scientist at Stanford University.
- Carried out a crucial measurement that serves as a foundation for many published physics findings.
- Acted as a referee, evaluating manuscripts submitted to Physical Review scientific journals.

Additional Roles

- Developer of a Plasma Panel Radiation Detector, Tel Aviv University (2011 – 2013)
- Member of The ATLAS Experiment, CERN Switzerland (2007 – 2013)
- Teaching Assistant, Tel Aviv University (2006 – 2013)

Education

- **Ph.D. in Physics**, Tel Aviv University, 2014
Thesis title: *Search for $\eta_c\pi^+\pi^-$ Resonances in Two-Photon Collisions at the BABAR Detector.*
Adviser: Prof. Soffer Abner.
- **M.Sc. in Physics**, Tel Aviv University, 2009
Thesis title: *Luminosity Measurement of the BABAR Experiment.*
Adviser: Prof. Soffer Abner.
- **B.Sc. in Physics**, Tel Aviv University, 2006
Graduated with distinction (Cum Laude).

Academic Publications

1. "Modeling noise in gravitational-wave observatories with transdimensional models." Guttman et al. Phys.Rev.D 111 (2025) 6, 063063
2. "A Gaussian process framework for testing general relativity with gravitational waves." Passenger et al. arXiv:2507.01294
3. "Transdimensional inference for gravitational-wave astronomy with Bilby." Tong et al. arXiv:2404.04460
4. "Possible Causes of False General Relativity Violations in Gravitational Wave Observations." Gupta et al arXiv:2405.02197
5. "Search for Resonances Decaying to $\eta_c\pi^+\pi^-$ in Two-Photon Interactions." The BABAR collaboration. Phys.Rev. D86 (2012) 092005.
6. "Time-Integrated Luminosity Recorded by the BABAR Detector at the PEP-II e^+e^- Collider." The BABAR collaboration. Nucl.Instrum.Meth. A726 (2013) 203-213

7. "Development of a Plasma Panel Radiation Detector: Recent Progress and Key Issues." Yiftah Silver et al. IEEE NSS 2011. arXiv:1205.6829
8. "The Detection of Ionizing Radiation by Plasma Panel Sensors: Cosmic Muons, Ion Beams and Cancer Therapy." Peter S. Friedman et al. SID Symposium, 2012. arXiv:1204.0677v1
9. "Plasma Panel Sensors for Particle and Beam Detection." Peter S. Friedman et al. Conference: C12-10-29, p.1775-1780 Proceedings arXiv:1211.5399

I am a co-author on all publications produced by the LIGO-Virgo-KAGRA collaboration since 2025, BABAR collaboration since 2009 and the ATLAS collaboration since 2007, representing more than 100 publications. The entire list is available upon request.

Patents

1. "Estimating color plane registration." Guttman, N., Sivan, Y., & Braverman, G. WO2015124196A1.
2. "Spectrophotometer diagnosis." Lena Gurevich, Nir Guttman, Ran Waidman WO2016174490A1.
3. "Adjusting imaging apparatuses." Nir Guttman, Gregory Braverman, Ran Waidman WO2017121483A1.
4. "Dynamic dot gain lookup table generation." Ran Waidman, Tal Frank, Nir Guttman WO2017182090A1.
5. "Printer with spectrophotometric measurement." Nir Guttman, Lena Gurevich, Gregory Braverman WO2017207013A1.
6. "Correcting reflectance output values." Nir Guttman, Ziv Berkovich, Gil Arie Uner WO2018068834A1.
7. "Spectrophotometric measurement estimation." Nir Guttman, Ziv Berkovich, Dan Ben Yaakov WO2019177590A1.
8. "Method and printer system for determining adjusted output settings." Nir Guttman, Gregory Braverman, Dan Ben Yaakov WO2019048029A1.
9. "Color calibration." Yoav Landau, Ran Waidman, Nir Guttman WO2021010998A1.

Non-Academic Publications

- INC, HP "Model-independent Spot Color Tone-Values Coverage Estimation", Technical Disclosure Commons (July 25, 2018).

Technical Skills

- **Programming Languages:** C++, C#, MATLAB, Python
- **Experienced with specific Python packages:** Numpy, Scikit-learn, Pandas, Keras, Matplotlib
- **Version control:** Git
- **Operating system:** Windows, Linux